The maximum-run enumeration of flat Fubini rankings

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Task 1: Resolve MathDB 367940. Let $s(n, r)$ count Fubini rankings of length n whose maximal strictly increasing runs have strictly increasing first entries, with exactly r runs. We prove the maximum-run enumeration proposed in Section 4.1 of Barrese, Elder, Harris, and Simpson [1]. Their displayed equation has indexing errors, which we make explicit before proving the intended enumeration for every size.

Use the second-order Eulerian convention of the cited OEIS entry A340556 [2]:

$$E_2(0,0) = 1, \qquad E_2(j,k) = kE_2(j-1,k) + (2j-k)E_2(j-1,k-1), \tag{1}$$

with entries outside $0 \le k \le j$ equal to zero. Write $\mathcal{E}_j(z) = \sum_{k=0}^{j} E_2(j,k)z^k$. For $j \ge 1$, this counts descents of Stirling permutations *including the final descent to an appended zero.*

Theorem 1. *The largest possible number of runs in a flattened Fubini ranking of length $2j+1$ is $j+1$. For every $j \ge 0$,*

$$\boxed{s(2j+1, j+1) = \mathcal{E}_j(2) = \sum_{k=0}^{j} E_2(j,k)2^k.} \tag{2}$$

More precisely, the number of these rankings with exactly $k+1$ distinct ranks is

$$N_j(k) = \sum_{m=0}^{k} (-1)^{k-m} \binom{k}{m} e_j(1, 2, \ldots, m), \qquad 0 \le k \le 2j, \tag{3}$$

where e_j is the elementary symmetric polynomial, with $e_0 = 1$ and $e_j(1, \ldots, m) = 0$ for $j > m$.

The indexing issue. The source's Conjecture 4.4, equation (4.1) in the publisher's current accepted manuscript, prints

$$s(2j+1, j-1) = \sum_{i=0}^{j-1} E_2(2j+1, i)2^i.$$

With its cited A340556 convention this literal formula is false: at $j = 2$ its left side is $s(5,1) = 1$, whereas its right side is 2. The surrounding text instead specifies odd length and the *largest possible* number of runs, and its Table 2 gives

$$s(3,2) = 2, \quad s(5,3) = 10, \quad s(7,4) = 82, \quad s(9,5) = 938.$$

It identifies this diagonal with A112487, whose defining formula is $\mathcal{E}_j(2)$ [3]. Thus (2) is precisely the enumeration specified by that text, table, and sequence. The proof below resolves that enumeration; noticing the printed mismatch alone is not the result.

1. Replace ranks by their order. A *packed word* uses exactly an alphabet $[q] = \{1, \dots, q\}$. There is a bijection from Fubini rankings to packed words: replace the smallest occurring rank by 1, the next by 2, and so on. Conversely, if letter i has multiplicity c_i, replace every i by

$$1 + c_1 + \cdots + c_{i-1}.$$

These are exactly the valid competition ranks with ties. Both maps preserve strict comparisons and equalities, hence preserve all the run conditions.

No run except the last can have length one: its single entry would be smaller than the next run's first entry, so the two would belong to the same increasing run. Consequently a word with r runs has length at least $2r-1$. At length $2j+1$, equality with $r = j+1$ forces the run lengths to be $2, 2, \dots, 2, 1$. Such a packed word has the form

$$a_0\, b_1 \mid a_1\, b_2 \mid \cdots \mid a_{j-1}\, b_j \mid a_j, \qquad a_0 < a_1 < \cdots < a_j, \quad b_i \geq a_i. \tag{4}$$

Conversely these inequalities make the displayed bars exactly the run boundaries. They also force a_0 to be the unique smallest letter.

Remove a_0 and decrease all remaining packed labels by one. The resulting objects are precisely the surjective assignments to $2j$ named positions satisfying

$$1 \leq a_1 < \cdots < a_j \leq k, \qquad a_i \leq b_i \leq k. \tag{5}$$

Any such assignment extends uniquely back to (4) by increasing all labels and adjoining the new minimum. For example, choosing $a_i = b_i = i$ gives a valid object for every j, so the run bound is attained. The case $j = 0$ is the unique one-letter ranking.

2. Count assignments before requiring surjectivity. Let $T_j(m)$ count assignments satisfying (5) into $[m]$, now allowing unused labels. After choosing the increasing sequence of a_i, each b_i has $m - a_i + 1$ choices. Therefore

$$T_j(m) = \sum_{1 \leq a_1 < \cdots < a_j \leq m} \prod_{i=1}^{j} (m - a_i + 1) = e_j(1, 2, \dots, m). \tag{6}$$

Choosing the used k-element subalphabet and packing it gives

$$T_j(m) = \sum_{k=0}^{2j} \binom{m}{k} N_j(k). \tag{7}$$

Binomial inversion proves (3). Moreover, the identity

$$\sum_{m=k}^{\infty} \frac{\binom{m}{k}}{2^{m+1}} = 1$$

gives the useful averaging formula

$$s(2j+1, j+1) = \sum_{k=0}^{2j} N_j(k) = \frac{1}{2} \sum_{m=0}^{\infty} e_j(1, 2, \dots, m)\, 2^{-m}. \tag{8}$$

There is no convergence issue: for fixed j the symmetric polynomial is bounded by a constant times m^{2j}.

3. Evaluate the average. We give the generating-function calculation explicitly. Put

$$H_j(t) = \sum_{m \geq 0} e_j(1, 2, \ldots, m) t^m, \qquad |t| < 1.$$

Adjoining the largest argument m in an elementary symmetric polynomial gives

$$e_j(1, \ldots, m) = e_j(1, \ldots, m-1) + m e_{j-1}(1, \ldots, m-1).$$

Thus $H_0(t) = (1-t)^{-1}$ and, for $j \geq 1$,

$$H_j(t) = \frac{t}{1-t} \frac{d}{dt}\big(t H_{j-1}(t)\big). \tag{9}$$

If $H_j(t) = t^j Q_j(t)/(1-t)^{2j+1}$, this recurrence becomes

$$Q_0(t) = 1, \qquad Q_j(t) = \big(j + (j-1)t\big) Q_{j-1}(t) + t(1-t) Q'_{j-1}(t). \tag{10}$$

For completeness, the Eulerian recurrence (1) follows by inserting the adjacent largest pair jj in a Stirling permutation of order $j-1$. There are $2j-1$ insertion gaps. If the old permutation has h descents, including the terminal descent, exactly h gaps preserve the count, and the other gaps increase it by one. This also covers $j = 1$, starting from the empty word. In polynomial form the recurrence is

$$\mathcal{E}_j(z) = (2j-1) z \mathcal{E}_{j-1}(z) + z(1-z) \mathcal{E}'_{j-1}(z).$$

Substitution shows that $t^j \mathcal{E}_j(1/t)$ satisfies (10) with the same initial value. Hence

$$Q_j(t) = t^j \mathcal{E}_j(1/t), \qquad H_j(t) = \frac{t^{2j} \mathcal{E}_j(1/t)}{(1-t)^{2j+1}}.$$

Finally (8) gives

$$s(2j+1, j+1) = \frac{1}{2} H_j(1/2) = \mathcal{E}_j(2),$$

as asserted. This proves the formula and the refinement for every $j \geq 0$. □

Verification and scope. The exact verifier passes 120 checks, including independent enumeration of all packed words through length 7, the conversion back to valid Fubini rankings, the refined counts, and rational generating-function identities. The first diagonal values are

$$1,\ 2,\ 10,\ 82,\ 938,\ 13778,\ 247210,\ 5240338,\ 128149802.$$

The proof, rather than these finite checks, establishes the infinite identity. The general two-variable enumeration and the source's separate total-count conjecture are not asserted here. The dated status audit located no matching full public claim and confirmed that the current MathDB record remains open.

Task 2: Construct a direct bijection to decorated Stirling permutations. Replace the averaging argument by a bijection between maximum-run rankings and Stirling permutations in which each descent, including the terminal descent, receives one of two colors.

Task 3: Enumerate the maximum-run rankings of even length. Determine $s(2j, j)$, keeping track of the possible run-length profiles and the number of distinct ranks.

Task 4: Enumerate all run counts. Extend the packed-word description to derive a recurrence or generating function for $s(n, r)$ for arbitrary n, r, and compare its total count with the source's separate generating-function conjecture.

Task 5: Determine the distribution of distinct ranks. Choose uniformly from the maximum-run rankings of length $2j + 1$. Use (3) to find the asymptotic mean, variance, and limiting distribution for the number of distinct ranks.

References

[1] K. Barrese, J. Elder, P. E. Harris, and A. Simpson, *Enumerating flat Fubini rankings*, The Art of Discrete and Applied Mathematics, accepted manuscript published 25 May 2026, DOI 10.26493/2590-9770.1883.9e1. Conjecture 4.4 and Table 2, page 19. https://adam-journal.eu/index.php/ADAM/article/view/1883. Author preprint: https://arxiv.org/abs/2504.06466.

[2] The OEIS Foundation, A340556, second-order Eulerian numbers, including the terminal descent. https://oeis.org/A340556.

[3] The OEIS Foundation, A112487, values $\mathcal{E}_j(2)$. https://oeis.org/A112487.